



AGGREGATED AGGREGATES

E-COMMERCE PLATFORM

Platform Build & Delivery Documentation

Wireframes, As-Built Diagrams & Module Status — A Specialised Operating Division of Besbpo Group

September 2026

EXECUTIVE SUMMARY

What This Document Is

This is the platform build documentation — distinct from, and later than, the earlier "Technical & Product Specification v1.0" (aggregated-aggregates-ecommerce-spec.pdf), which was the pre-build planning document. That spec fixed scope and architecture before any code existed. This document covers what was actually built: the real repository structure, the modules that exist and run, and the honest list of what remains open.

It contains three kinds of visual: the five original Phase 1 planning wireframes (low-fidelity, produced before the build started, and still a reasonably accurate map of page structure); new as-built diagrams (system architecture, data model, payment routing) generated directly from the real repository — folder names, module names, and table names below are copied verbatim from the code, not reconstructed from memory; and updated page wireframes that reflect what the build actually added beyond the original plan (the ad system's PromoSlot, the technical-specification tab, labelled form fields, etc.).

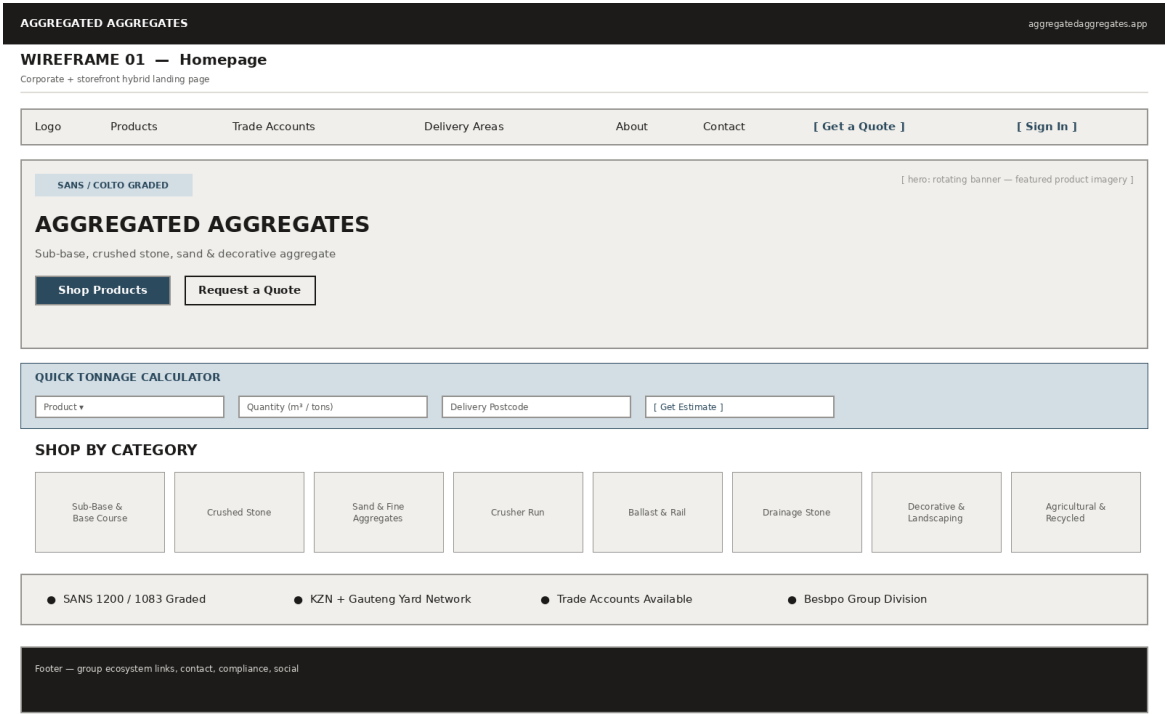
A note on how these diagrams were produced: this build was carried out in a sandboxed environment with no ability to run "next build" and take real screenshots (outbound npm/package-registry access is blocked here — see README.md). Every diagram in this document is therefore a schematic box diagram, not a screenshot, generated to accurately reflect the real component names, real file paths, and real data shapes in the repository. Where a figure or label appears in a diagram, it is drawn from the actual source file named alongside it.

Relationship to the Original Spec

- The original spec's Feature Adoption Matrix, Payment Strategy, and Fulfilment sections are still the source of truth for high-level scope decisions — this document does not repeat them.
- The original spec's 6 "Aggregated-Aggregates-Specific Modules" have all been built, plus additional modules the spec didn't originally cover: WhatsApp Commerce, Instagram/Facebook catalogue feed, the scaled-down ad system, and a blog/CMS foundation.
- Where this document's diagrams differ from the original spec's wireframes (e.g. the product detail page now has Typical Uses and Handling & Storage tabs, not just Specification/Grading Curve/Compliance/Delivery), that reflects real scope added during the build, confirmed with Fortune along the way.

PHASE 1 — PLANNING**Original Planning Wireframes (Pre-Build)**

Produced during Phase 1 planning, before any code was written. Included here for continuity with the original spec document. Superseded in detail (not in intent) by the "As-Built Page Structure" section further on.



Original Wireframe 1 — Homepage: corporate + storefront hybrid, quick tonnage calculator above the fold.

Filterable grid with unit-of-sale toggle

[Home](#) / [Products](#) / [Crushed Stone Aggregates](#)

FILTERS

Category ▾

Grading / Size ▾

Unit of Sale (Bulk / Bag) ▾

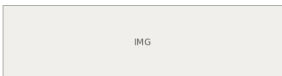
Segment (Civil/Comm/Ind/Res) ▾

Delivery Radius ▾

Apply Filters

48 products

Sort: Relevance ▾



G5 Natural Gravel

R400.00 /ton • R580.00 /m³

[Bulk / Bag ▾](#) [Add to Quote](#)



19mm Crushed Stone (Dolomite)

R400.00 /ton • R580.00 /m³

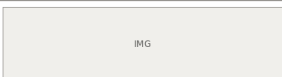
[Bulk / Bag ▾](#) [Add to Quote](#)



River Sand (Washed)

R400.00 /ton • R580.00 /m³

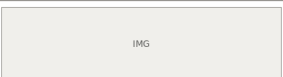
[Bulk / Bag ▾](#) [Add to Quote](#)



Crusher Run 0-19mm

R400.00 /ton • R580.00 /m³

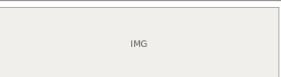
[Bulk / Bag ▾](#) [Add to Quote](#)



Rail Ballast

R400.00 /ton • R580.00 /m³

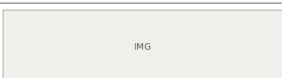
[Bulk / Bag ▾](#) [Add to Quote](#)



French Drain Stone

R400.00 /ton • R580.00 /m³

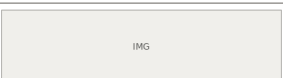
[Bulk / Bag ▾](#) [Add to Quote](#)



River Pebble

R400.00 /ton • R580.00 /m³

[Bulk / Bag ▾](#) [Add to Quote](#)



Agricultural Lime

R400.00 /ton • R580.00 /m³

[Bulk / Bag ▾](#) [Add to Quote](#)



Recycled Crushed Concrete (RCA)

R400.00 /ton • R580.00 /m³

[Bulk / Bag ▾](#) [Add to Quote](#)

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AGGREGATED AGGREGATES

aggregatedaggregates.app

WIREFRAME 03 — Product Detail — Bulk/Bag Calculator

Unit toggle drives live price + delivery-band estimate

Home / Products / Crushed Stone Aggregates / 19mm Crushed Stone (Dolomite)

PRODUCT IMAGE GALLERY

thumb

thumb

thumb

thumb

19mm Crushed Stone (Dolomite)

Crushed Stone Aggregates • SANS 1083

R400.00 / ton

R580.00 / m³ • R11.78 / 20kg bag

TONNAGE / BAG CALCULATOR

Unit of Sale

☒ Bulk (ton/m³)

☐ Bagged

Quantity

[10] m³ ⇄ [14.5] tons

Customer Tier

Retail ▾ (sign in for Trade/Volume pricing)

Delivery Distance

[Postcode/Address] → 32km band

Subtotal: R5,800.00 + Delivery: R450.00

Add to Quote

Request Bulk RFQ

Specification

Grading Curve

Compliance Docs (SANS/COA)

Delivery & Returns

Tab content — grading curve chart / SANS 1083 spec table

Original Wireframe 3 — Product detail: the tonnage/bag calculator, live-priced against tier and delivery distance.

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WIREFRAME 04 — RFQ / Bulk Quote Request

Multi-step flow for civil bulk & >100km delivery quotes

1. Project Details

2. Materials & Quantities

3. Delivery Location

4. Review & Submit

STEP 2 — Materials & Quantities

Product	Grading/Size	Unit	Quantity	Est. Unit Price	
G3 Crushed Stone	G3 (COLTO)	m³	120	R580.00	×
13.2mm Crushed Stone	13.2mm	ton	45	R400.00	×
Ferrocrite Ballast	Graded blend	ton	80	R256.20	×

+ Add Another Product

Note: orders ≥ 10m³ (Volume/Civil Bulk tier) or delivery > 100km are quoted individually — our team responds within 1 business day with delivered pricing.

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Original Wireframe 4 — RFQ / bulk quote request: the path for Volume-tier and >100km orders.

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WIREFRAME 05 — Trade Account Dashboard

Customer portal — tier status, orders, standing agreements

Dashboard

Order History

Quotes / RFQs

Standing Agreements

Delivery Addresses

Invoices & Statements

Account Settings

Welcome back — Company: XYZ Civil Contractors (Pty) Ltd

ACCOUNT TIER

Contractor / Trade
(8% off list)

ORDERS THIS MONTH

6 orders

OUTSTANDING BALANCE

R48,320.00

STANDING AGREEMENT

KZN Civil — active

RECENT ORDERS

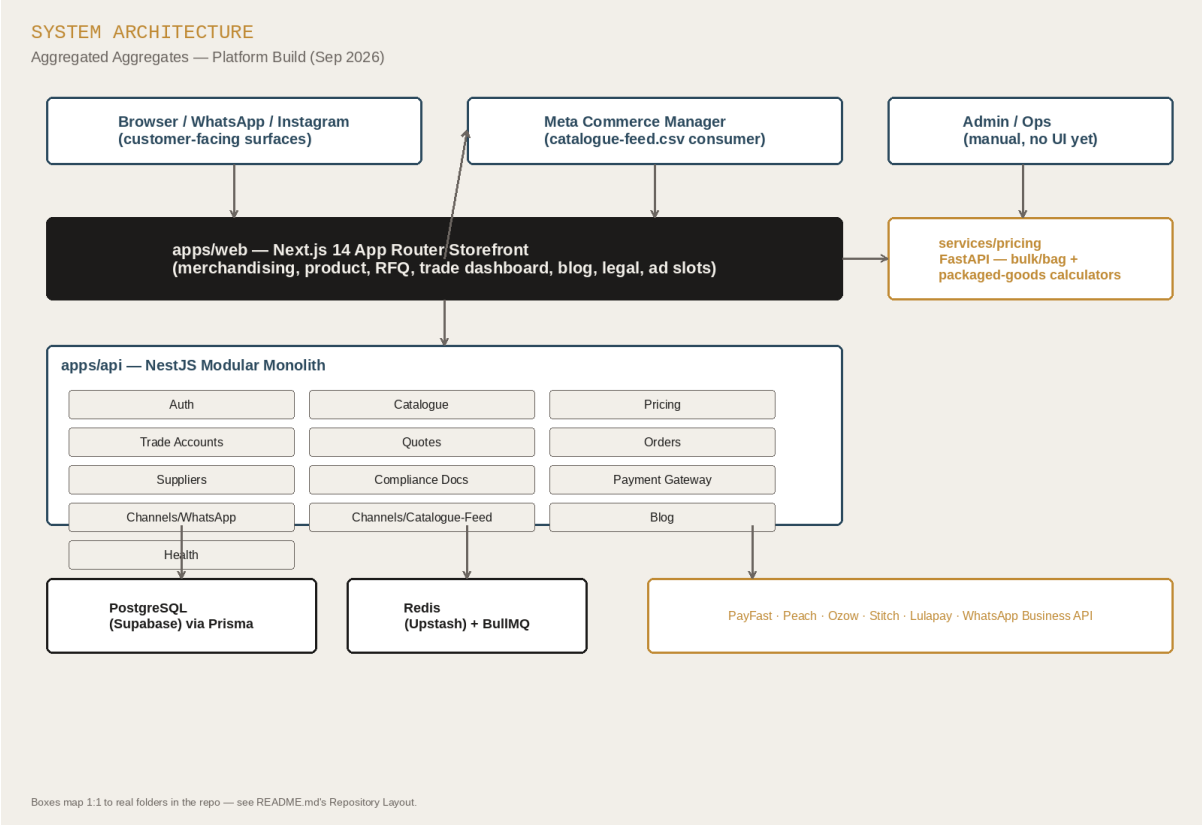
Order #	Date	Products	Delivery Yard	Status	Total
AA-10245	18 Sep 2026	G5 Gravel, 19mm Stone	Pinetown Yard	Delivered	R14,200
AA-10238	11 Sep 2026	River Sand x 10m³	Pinetown Yard	In Translt	R5,555
AA-10221	29 Aug 2026	Crusher Run 0-19mm	Germiston Yard	Delivered	R8,960

Original Wireframe 5 — Trade account dashboard: tier status, recent orders, standing agreements.

AS - BUILT

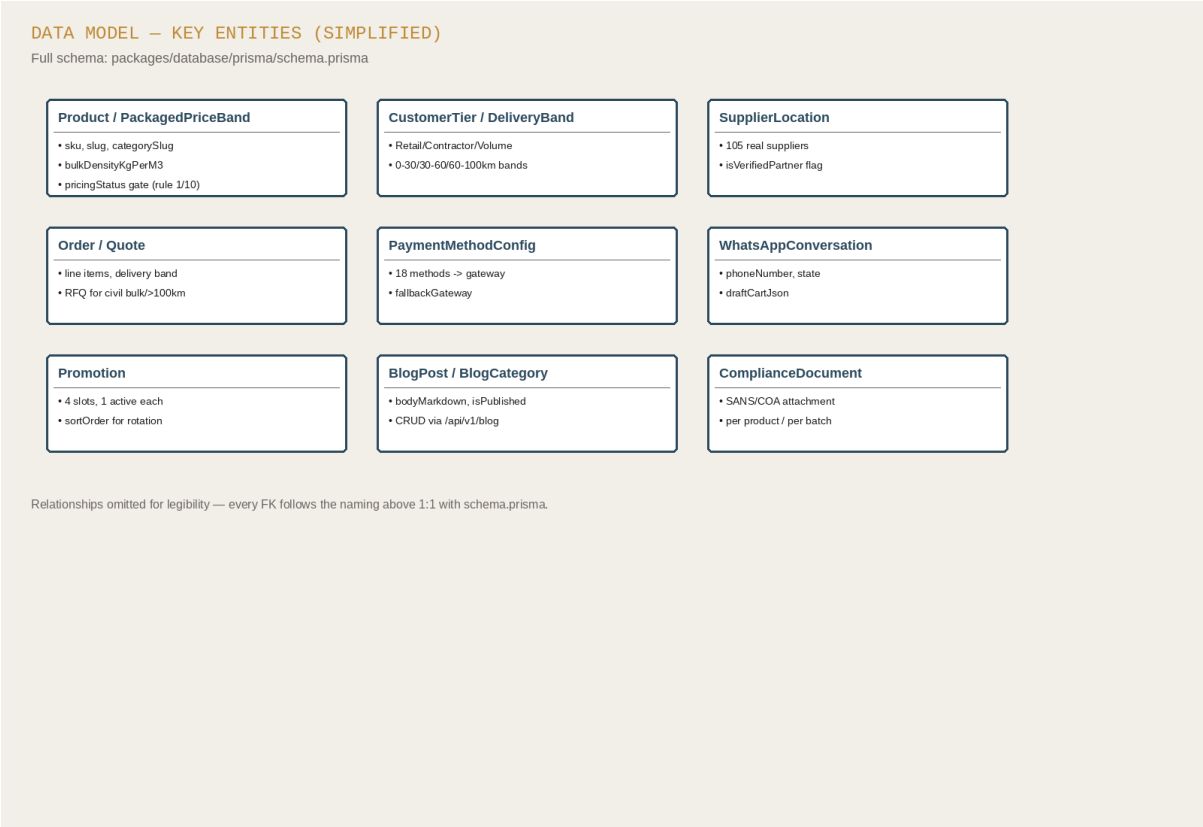
System Architecture (As Built)

Every box below is a real folder or module in the delivered repository — see README.md's "Repository Layout" for the same structure in text form.



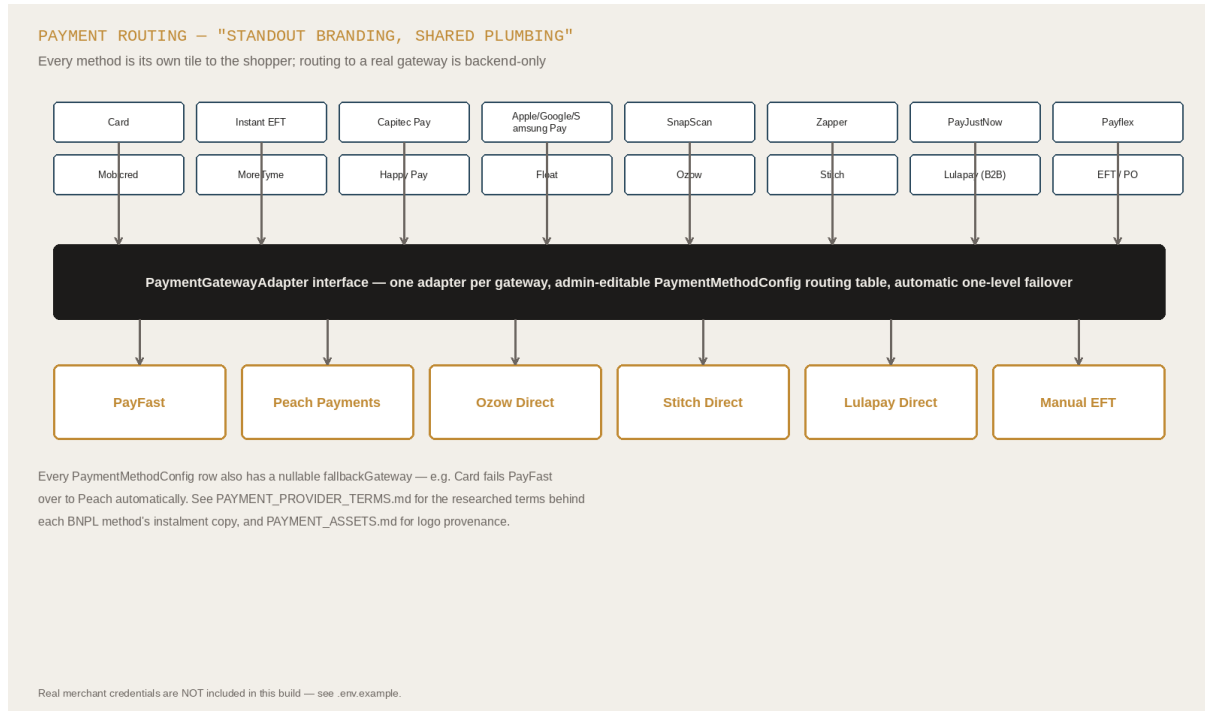
Data Model (As Built, Simplified)

The full schema is `packages/database/prisma/schema.prisma`. This simplified view groups the ~25 real Prisma models into 9 logical clusters for legibility.



Payment Routing (As Built)

The full 18-method payment lineup, each a standout tile to the shopper, routed backend-only through a shared PaymentGatewayAdapter interface. See `apps/api/src/payment-gateway/` for the real adapter implementations.



AS-BUILT**As-Built Page Structure (Updated Wireframes)**

These reflect the real component import order and content in the delivered Next.js pages — including scope added since the original Phase 1 wireframes (the ad system's PromoSlot placements, the WhatsApp order button, the expanded product-detail tab set, and labelled/accessible form fields).

WIREFRAME — HOMEPAGE (apps/web/src/app/page.tsx)

HeroBanner + HeroSlideshow

10 sourced Unsplash images, auto-rotating

ComplianceBar / TrustBadges

SANS/COLTO badges, trust strip

CategoryGrid

9 product categories

BulkVsBagged

Explainer: ton/m³ vs bag pricing

FeaturedProducts

Sample SKUs, live price calc

PromoSlot — HOMEPAGE_SECONDARY_BANNER

Ad system: civil/road-works creative

IndustriesWeServe

Civil / commercial / residential / agri

B2BBulkSection

Cement, binders, admixtures

QuarryToSite / TierComparison / CoverageStrip

Process + tier + delivery-area strips

WhatsAppCta

wa.me/27683676276 order-via-chat

PromoStrip — trade account CTA

Static banner, 8%/15% off

Sections top-to-bottom match the real component import order in page.tsx

As-Built — Homepage (apps/web/src/app/page.tsx). Now includes a live PromoSlot (HOMEPAGE_SECONDARY_BANNER) between Featured Products and Industries We Serve.

WIREFRAME — PRODUCT DETAIL (apps/web/src/app/products/[slug]/page.tsx)

Breadcrumb: Home / Products / {Category} / {Product}

PRODUCT IMAGE GALLERY

Category · Grading standard

Product name (H1)

Category description

BulkBagCalculator / PackagedUnitSelector

DeliveryEstimator

WhatsAppOrderButton + SocialShareButtons

Specification | Typical Uses | Handling & Storage | Compliance Docs | Delivery & Returns

Technical Specification table:

• Grading standard

• Particle size range

• Bulk density (kg/m³)

• Available units

• Typical uses (bulleted)

• Handling & storage notes

Product schema (schema.org/Product) JSON-LD emitted for SEO.

As-Built — Product Detail (apps/web/src/app/products/[slug]/page.tsx). Expanded to 5 tabs including new Typical Uses and Handling & Storage sections, plus schema.org/Product JSON-LD.

WIREFRAME — RFQ / BULK QUOTE FLOW (apps/web/src/app/quote/page.tsx)

1. Project Details

2. Materials & Quantities

3. Delivery Location

4. Review & Submit

PromoSlot — QUOTE_FLOW_UPSELL
"Open a Trade Account for standing pricing"

Active step content (Step 1 shown):

Project / site name [labelled input]

Company name [labelled input]

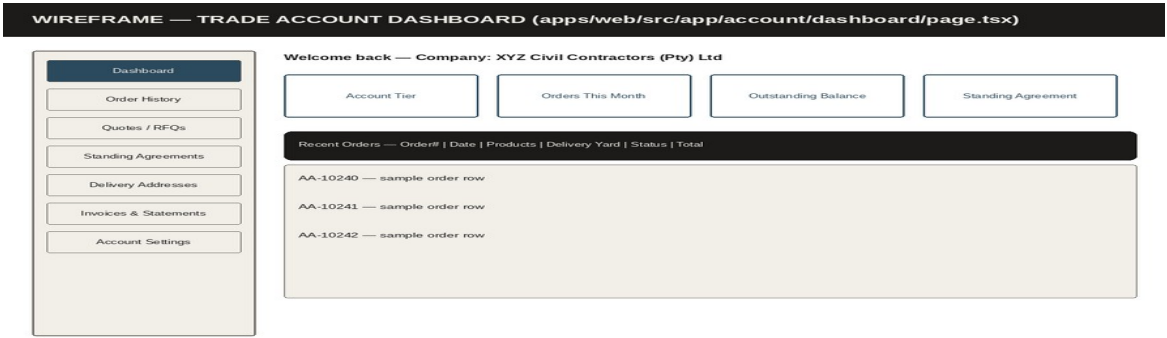
Contact email [labelled input]

— Back

Continue —>

4-step client component: civil-bulk (>=10m3) and >100km deliveries route here

As-Built — RFQ / Bulk Quote Flow (apps/web/src/app/quote/page.tsx). All form fields are now properly labelled (accessibility fix), and carries a QUOTE_FLOW_UPSELL promo slot.



As-Built — Trade Account Dashboard (apps/web/src/app/account/dashboard/page.tsx).

DELIVERY STATUS

Module & Feature Build Status

An honest, itemised status for every module discussed across the build — not just the ones that went smoothly.

Module	Status	Notes
Core storefront (merchandising, product, category)	Built	Next.js App Router, ~10 pages, real component wiring.
Bulk/bag calculator + distance-banded delivery	Built	FastAPI pricing microservice, cross-checked calculators.
Trade accounts, RFQ/quote flow	Built	3-tier model; 4-step client-side RFQ flow.
Payment gateway (18 methods, 6 adapters)	Built	PayFast/Peach/Ozow/Stitch/Lulapay/Manual EFT adapters + failover; no real merchant credentials included (.env.example).
WhatsApp Commerce	Built	apps/api/src/channels/whatsapp — conversation state machine, draft cart, order handoff.
Instagram/Facebook catalogue feed	Built	GET /channels/catalogue-feed.csv — Meta Commerce Manager-compatible.
Ad system (Promotion model + PromoSlot)	Built & wired live	4 slots (homepage/category/quote/footer), each with 1 active sourced-photo creative, seeded in seed.ts.
Hero slideshow imagery	10 of 10-12 sourced	Real, licensed Unsplash photography — see media.ts for full provenance and the 1-2 remaining slots' status.
Blog / CMS	Backend built; admin UI + auth guard NOT built	Real NestJS CRUD module + Prisma models; storefront still reads hardcoded data.ts. See BLOG_CMS.md.
Corporate contact details & address	Built	Real phone lines (1 live, 3 RICA-pending), registered address — in footer, /contact, and Organization JSON-LD.
Payment & social icon assets	Placeholders only	26 generated placeholder SVGs at the correct file paths — pending real Brandfetch assets (see PAYMENT_ASSETS.md).
Real supplier network on storefront	Built	All 105 researched suppliers (87 verified partner + 18 B2B-bulk leads), grouped by province, with address text — apps/web/src/data/suppliers.ts.
Product technical specifications	Built	Particle size range, typical uses, handling/storage notes added to all 17 SKUs; rendered as a spec table + JSON-LD.
SEO & platform best practices	Built	Metadata on every page, error/global-error/loading boundaries, Organization + Product + BlogPosting JSON-LD, Helmet + /health endpoint.

Module	Status	Notes
This PDF (wireframes, mockups, diagrams)	Built	This document.

BEFORE LAUNCH

Open Items — What Still Needs Human Action

Carried over honestly rather than glossed over. None of these block continued development; they block a public launch.

Payments & Compliance

- Real merchant/gateway credentials (PayFast, Peach, Ozow, Stitch, Lulapay) are not included — see `.env.example`.
- The 3 geographic landlines (Johannesburg/Cape Town/Durban) are paid for and assigned but RICA-compliance activation is still pending.

Assets

- Every payment-method logo and social-media icon is a generated placeholder SVG, not the real brand asset — Fortune's own words: "I'll fetch everything on Brandfetch." Swapping them in is a same-filename drop-in (see `PAYMENT_ASSETS.md`).
- 1-2 of the 10-12 planned hero slideshow images remain to be sourced (Unsplash rate-limited further lookups); the method to finish this is documented in `media.ts`.

Blog / CMS

- No admin UI exists yet for writing posts — only a working CRUD API. See `BLOG_CMS.md` for the recommended next steps.
- The blog API's write routes (create/update/delete) are not yet behind an auth guard — this matches the rest of the platform's current auth maturity (Phase 2 SSO guard also not yet wired), but must be fixed before public deployment.

Data & Verification

- Supplier addresses are free-text (from real research), not yet geocoded to lat/lng — the delivery-distance calculator currently uses representative distances, not these exact coordinates.
- This build's environment could not run npm/pip installs (registry access blocked) — dependency-bearing code was verified by manual review, bracket-balance and import-path checks, and direct execution of dependency-free logic, not by a real compile/test run. Run a full ``npm install && npm run build`` in an environment with registry access before deploying.

Pricing

- Every price on the platform traces to a named benchmark or is explicitly gated as provisional/quote-only/blocked (`AGENTIC_RULES.md` rule 1/10) — reconcile all "Ready — benchmarked" figures against the live pricing engine before go-live.